Shayan Dodge

PhD Researcher in Computational Electromagnetics and Scientific Machine Learning
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ORCID Google Scholar LinkedIn Website GitHub

Research Profile

PhD researcher in Computational Electromagnetics and Scientific Machine Learning with experience applying AI to electromagnetic simulation, power and energy systems, bioelectromagnetics, inverse problems, and scientific computing. Developed physics-informed neural networks, neural operators, generative AI models, and deep learning frameworks for solving PDEs, surrogate modeling, signal analysis, optimization, and simulation acceleration.

Research Highlights

- 18 peer-reviewed publications: 15 journal articles and 3 conference papers.
- Research experience in Scientific Machine Learning, PINNs, Neural Operators, Computational Electromagnetics, and AI for Physics.
- Student Editorial Assistant, *IEEE Journal on Multiscale and Multiphysics Computational Techniques*.
- Reviewer for 7 international journals.
- Recipient of a competitive research grant for "Application of Machine Learning to Electromagnetism Problems," University of Pisa, 2024. Funded through the FELINES and STEM-DEEP projects.

Education

Ph.D. in Electrical Engineering Oct 2024 – Present
University of Pisa (DESTEC), Italy

M.Sc. in Plasma Physics Sep 2017 – Feb 2020
Shahid Beheshti University, Laser and Plasma Research Institute, Iran
GPA: 18.27/20 (WES evaluated as U.S. GPA 3.87/4.00)

B.Sc. in Physics Oct 2013 – Jun 2017
Kharazmi University, Department of Physics, Iran
GPA: 18.46/20 (WES evaluated as U.S. GPA 3.98/4.00)

Research Experience

Research Fellow – University of Pisa (DESTEC) Jan 2024 – Jan 2026

Worked on AI-driven computational electromagnetics projects within the PRIN-funded *FELINES* and *STEM-DEEP* research programs.

Teaching and Student Supervision

Teaching Assistant

- **Electrical Circuit Theory Course** 2025/2026
  University of Pisa, Italy, Department of Information Engineering, B.Sc. in Electrical Engineering
- **Electrical Circuit Theory Course** 2024/2025
  University of Pisa, Italy, Department of Information Engineering, B.Sc. in Electrical Engineering

B.Sc. Thesis Co-Supervisor

- **Deep Learning for TMS using FEM-Generated Datasets** 2025/2026
  University of Pisa, Italy, B.Sc. in Biomedical Engineering
- **Automated TMS Dataset Generation using MATLAB and SimNIBS** 2024/2025
  University of Pisa, Italy, B.Sc. in Biomedical Engineering

Research Interests

- Scientific Machine Learning and AI for Physics.
- Physics-Informed Neural Networks and Variational PINNs.
- Neural Operators, DeepONet, and Surrogate Modeling.
- Computational and Applied Electromagnetics.
- Inverse Problems and Optimization.
- Geophysical Electromagnetic Imaging and Electrical Resistivity Tomography (ERT)
- Power and Energy Systems.
- Bioelectromagnetics, TMS, SAR, MRI, and EIT.
- High-Performance Scientific Computing.

Funded Research Projects

AIGEO – AI for Electromagnetic Geophysical Inversion Aug 2026 – Present
Partners: University of Pisa (DESTEC); Geostudi Astier (geophysical engineering company)
Funding: Regione Toscana [Website]

- Developing data-driven and physics-informed neural networks for electromagnetic geophysical inverse problems, with applications to Electrical Resistivity Tomography (ERT).

FELINES – AI for Lightning Detection and Power-System Protection Jan 2024 – Jan 2026
Partners: University of Pisa; University of Campania; University of Genoa
Funding: PRIN (Research Projects of National Interest), Italian Ministry of University and Research (MUR) [website]

- Developed deep-learning methods for lightning localization and peak-current estimation. [Link]
- Developed ML-based approaches for predicting lightning-induced overvoltages and related inverse problems. [Link]
- Contributed to computational and data-driven methodologies for power-system protection.

STEM-DEEP – Deep Learning for Personalized TMS Jan 2024 – Jan 2026
Partners: University of Pisa; University of Pavia; Politecnico di Milano
Funding: PRIN (Research Projects of National Interest), Italian Ministry of University and Research (MUR) [website]

- Developed Generative AI models for personalized transcranial magnetic stimulation (TMS), combining VAE-based electric-field generation with CNN-based inverse modeling to optimize coil positioning and stimulation parameters. [Link]

Other Research Projects & Collaborations

Operator Learning for Parametric Electromagnetic Modeling Apr 2025 – Present
University of Pisa; KU Leuven, Belgium

- Developed *neural-operator* surrogate models for accelerated simulation of electromagnetic devices.

Physics-Informed Neural Networks for Electromagnetic PDEs Jul 2024 – Present
University of Pisa; University of Campania

- Developed *INI-VPINN* formulations inspired by FEM for multi-material and inverse electromagnetic problems.
- Developed *STAR-PINN* architectures using stacked adaptive residual learning for nonlinear electromagnetic simulations.
- Developed *hybrid BEM–PINN* frameworks integrating Boundary Element Methods (BEM) with physics-informed neural networks for electromagnetic PDEs.

Deep Learning for Real-Time SAR Assessment Sep 2024 – Mar 2025
University of Pisa; University of Pavia

- Developed deep-learning surrogate models for *SAR hot-spot* prediction in wearable electromagnetic devices, enabling rapid exposure assessment.

GPU-Accelerated FDTD for Microwave and Plasma Applications Sep 2018–Jan 2024
Shahid Beheshti University

- Investigated *microwave hyperthermia* using analytical and FDTD models to characterize electromagnetic energy deposition and temperature distributions in biological tissues.
- Developed *GPU-accelerated FDTD* methods for electromagnetic wave propagation in plasma, leveraging parallel computing to achieve up to a 20× speedup over serial implementations.

Technical Skills

Machine Learning and AI
PyTorch, TensorFlow, JAX, Deep Learning, Neural Networks, PINNs, Neural Operators, DeepONet, Scientific Machine Learning, Surrogate Modeling

Scientific Computing
Computational Electromagnetics, FEM, FDTD, BEM, Inverse Problems, Numerical PDEs, Signal Processing

Programming
Python, MATLAB, C++, CUDA

High-Performance Computing
GPU Computing, CUDA Acceleration, Parallel Computing, Scientific Workflows

Simulation and Engineering Tools
ANSYS Maxwell, CST Studio Suite, FEniCS, Gmsh, SimNIBS

Software and Development
Git, Linux, Scientific Software Development

Selected Publications

1. **Dodge, S.**, Barmada, S., & Formisano, A. (2026). INI-VPINN: A Variational Physics-Informed Neural Network with Implicit Neumann and Interface Handling for Multi-Material Domains with Geometric Singularities. *Journal of Computational Physics*. [Paper] [GitHub] [LinkedIn]

2. **Dodge, S.**, Barmada, S., & Formisano, A. (2025). A STacked Adaptive Residual PINN (STAR-PINN) Approach to 2D Time-Domain Magnetic Diffusion in Nonlinear Materials. *IEEE Access*. [Paper] [GitHub] [LinkedIn]

3. **Dodge, S.**, Nicora, M., Barmada, S., et al. (2025). A deep learning based lightning location system. *Electric Power Systems Research*. [Paper] [GitHub] [LinkedIn]

4. Barmada, S., **Dodge, S.**, Tucci, M., et al. (2024). A Novel Hybrid Boundary Element–Physics Informed Neural Network Method for Numerical Solutions in Electromagnetics. *IEEE Access*. [Paper] [GitHub] [LinkedIn]

Journal Articles

1. **Dodge, S.**, Barmada, S., & Formisano, A. (2026). INI-VPINN: A Variational Physics-Informed Neural Network with Implicit Neumann and Interface Handling for Multi-Material Domains with Geometric Singularities. *Journal of Computational Physics*. [Link]

2. Toghranegar, S., **Dodge, S.**, Barmada, et al. (2026). Physics-Informed Deep Operator Networks for Parametric Modelling of Electromagnetic Devices: Magnetostatics to Magnetoquasistatics. *IEEE Transactions on Magnetics*. [Link]

3. Barmada, S., **Dodge, S.**, Formisano, A., Di Barba, P., & Mognaschi, M. E. (2026). PINN-Based Resolution of Inverse Non-linear Magnetostatic Problems. *COMPEL*. [Link]

4. Codecasa, L., Zhu, X., Di Rienzo, L., Barmada, S., **Dodge, S.**, et al. (2026). Fast Model Order Reduction Based Approach for Transcranial Magnetic Stimulation with Varying Coil Positioning. *IEEE Access*. [Link]

5. Formisano, A., **Dodge, S.**, & Barmada, S. (2025). A Comparison of Machine Learning and Classical Numerical Approaches for the Resolution of Electromagnetics Problems. *IET Science, Measurement & Technology*, 19(1), e70034. [Link]

6. Barmada, S., **Dodge, S.**, & Formisano, A. (2025). Weak Formulation for Physics-Informed Neural Networks in the Resolution of Analysis Problems in Electromagnetics. *IEEE Transactions on Magnetics*. [Link]

7. Cosentino, G., Zaffina, C., Zoccola, C., Fresia, M., Merli, S., Mauramati, S., Bertino, G., Todisco, M., **Dodge, S.**, et al. (2025). Unilateral EMG-Guided Botulinum Toxin for Retrograde Cricopharyngeus Dysfunction. *Toxins*, 17(9), 458. [Link]

8. Barmada, S., **Dodge, S.**, Brignone, M., et al. (2025). Relating transmission line overvoltages and lightning location: a machine learning-based procedure. *COMPEL*. [Link]

9. **Dodge, S.**, Barmada, S., & Formisano, A. (2025). A STacked Adaptive Residual PINN Approach to 2D Time-Domain Magnetic Diffusion in Nonlinear Materials. *IEEE Access*. [Link]

10. **Dodge, S.**, Fontana, N., Mognaschi, M. E., et al. (2025). A Deep Learning Based Prediction of Specific Absorption Rate Hot-Spots Induced by Broadband Electromagnetic Devices. *IET Science, Measurement & Technology*, 19(1), e70009. [Link]

11. **Dodge, S.**, Nicora, M., Barmada, S., et al. (2025). A deep learning based lightning location system. *Electric Power Systems Research*, 242, 111437. [Link]

12. Sekehravani, E. A., **Dodge, S.**, Barmada, S., et al. (2025). Preliminary Breakdown Pulses: A review on available data and models. *Electric Power Systems Research*, 242, 111463. [Link]

13. Barmada, S., **Dodge, S.**, Tucci, M., et al. (2024). A Novel Hybrid Boundary Element–Physics Informed Neural Network Method for Numerical Solutions in Electromagnetics. *IEEE Access*. [Link]

14. Niknam, A. R., **Dodge, S.**, Hajian, M., et al. (2024). Characterization of microwave heating for hyperthermia cancer treatment. *Waves in Random and Complex Media*, 34(1), 211–225. [Link]

15. **Dodge, S.**, Shafiee, M., & Shokri, B. (2022). Application of GPU-accelerated FDTD method to electromagnetic wave propagation in plasma using MATLAB Parallel Processing Toolbox. *arXiv preprint*. [Link]

Journal Articles Under Review

1. **Dodge, S.**, Mestriner, D., Nicora, M., Barmada, S., Brignone, M., Formisano, A., & Procopio, R. (2026). Prediction of lightning flashover on overhead distribution lines from preliminary breakdown voltages based on deep learning. *Electric Power Systems Research*. Under review. [Link]
2. Hajian, M., **Dodge, S.**, Fontana, N., & Barmada, S. (2026). Fast parametric prediction of coil coupling in wireless power transfer systems using physics-informed DeepONet. *IEEE Transactions on Magnetics*. Under review.
3. Barmada, S., **Dodge, S.**, & Formisano, A. (2026). Isoparametric mapped weak-form PINNs for arbitrary quadrilateral elements. *IEEE Transactions on Magnetics*. Under review.
4. **Dodge, S.**, Toghranegar, S., Formisano, A., Sabariego, R. V., & Barmada, S. (2026). Physics-Informed Fourier Neural Operators (PI-FNO) for Time-Domain Magnetoquasistatic Analysis of Nonlinear Electromagnetic Devices. *IEEE Journal on Multiscale and Multiphysics Computational Techniques*. Under review.
5. **Dodge, S.**, Fontana, N. Tucci, M, Barmada, S., et al. (2026). Real-Time Patient-Specific TMS Coil Placement Optimization Using a Generative Deep Learning Framework. *Scientific Reports*. Under review.

Conference Papers

1. **Dodge, S.**, Hajian, M., & Barmada, S. (2026). A non-conforming domain-decomposed variational physics-informed neural network for electromagnetic problems in complex geometries. *ACES*(pp. 1-2). IEEE. [Link]
2. Barmada, S., Bonfiglio, A., Brignone, M., **Dodge, S.**, Formisano, A., Procopio, R., & Akbari Sekehavani, E. (2025). A Survey of Measurements and Analyses of Preliminary Breakdown Pulses in Lightning Flashes. *EEEIC/I&CPS Europe*. [Link]
3. Nicora, M., Procopio, R., Brignone, M., **Dodge, S.**, et al. (2025). A Deep Learning Model for Lightning Location and Peak Current Estimation from Induced Overvoltages. *SIPDA*. [Link]

Academic Service

- Student Editorial Assistant, *IEEE Journal on Multiscale and Multiphysics Computational Techniques* (Dec 2025 – Present).
- Reviewer for:
  - *IEEE Transactions on Magnetics*
  - *IEEE Journal on Multiscale and Multiphysics Computational Techniques*
  - *COMPEL: The International Journal for Computation and Mathematics in Electrical and Electronic Engineering*
  - *Applied Computational Electromagnetics Society Journal*
  - *Physics of Plasmas*
  - *Optics Express*
  - *Waves in Random and Complex Media*

Honors and Awards

- Recipient of a competitive research grant, "Application of Machine Learning to Electromagnetism Problems," DESTEC, University of Pisa, 2024.
- Ranked 1st among Plasma Physics graduates, Shahid Beheshti University, 2020.
- Ranked 1st among all undergraduates of the Physics Department, Kharazmi University, 2017.
- Top 1.5% in Iran's BSc National Entrance Exam among 251,000 participants, 2013.

Professional Memberships

- IEEE Member
- IEEE Antennas and Propagation Society

Languages

- Persian: Native
- English: Professional working proficiency

Pisa, Italy, October 10, 2026



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